

MSE 312 — TB6568KQ Motor Driver + SN7400 NAND Bench Sheet

PWM speed + direction logic into IN1/IN2 · sourced from the TB6568KQ datasheet (I/O function table, p.6) and the SN7400 quad 2-input NAND

1. TB6568KQ pinout — as you hold it

View the printed **TOSHIBA** face with the metal heat tab behind and the pins pointing down. Pins number **1 (left) to 7 (right)**. Flipping to the tab side mirrors the order. The heat tab is internally tied to GND.

TB6568KQ — front view (HSIP7-P-2.54A)

"TOSHIBA" face toward you · heat tab behind · pins down · pin 1 = left



Pin	Name	Function
1	IN1	Control input 1 (logic, 5V)
2	IN2	Control input 2 (logic, 5V)
3	OUT1	Motor terminal A
4	GND	Ground (also heat tab)
5	OUT2	Motor terminal B
6	N.C.	No connect (creepage gap)
7	VM	Motor supply +, 10-45V typ (50V max)

2. Datasheet I/O function table (drives everything)

Straight from the TB6568KQ datasheet. Direction reads CW/CCW depending on which way the two motor leads are landed on OUT1/OUT2.

IN1	IN2	OUT1	OUT2	Mode
L	H	L	H	Drive one way
H	L	H	L	Drive other way
H	H	L	L	Short brake

L	L	Hi-Z	Hi-Z	Stop / coast
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3. The logic you need

Speed control chops a logic input, not the motor rail. Feed the driver **one PWM pin + two direction pins** and gate them:

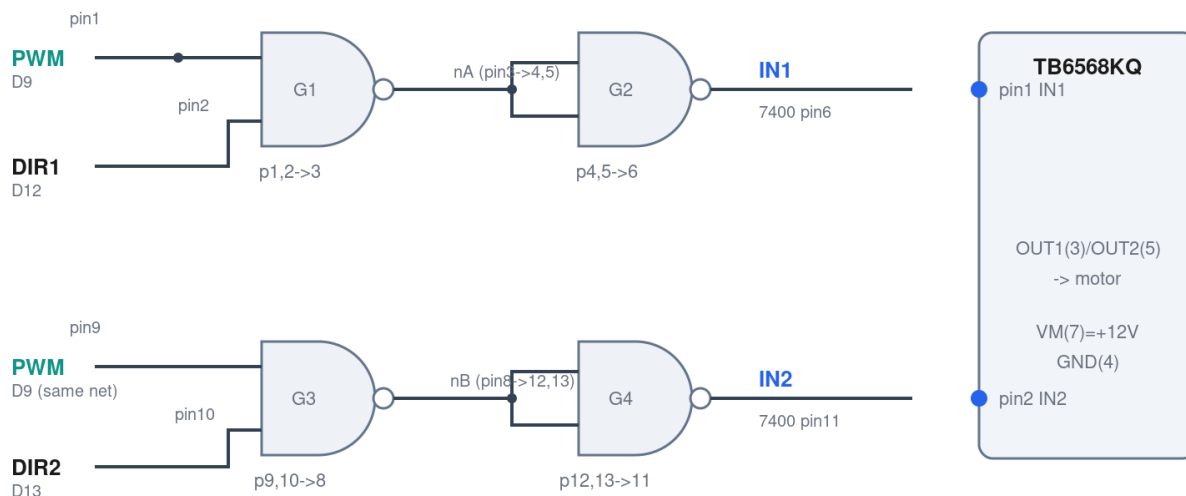
IN1 = PWM AND DIR1

IN2 = PWM AND DIR2

A NAND with both inputs tied together is an inverter, so AND = NAND then invert = two gates. Two ANDs use **exactly 4 gates = one SN7400**.

4. Gate wiring (one SN7400)

One SN7400 to generate IN1 / IN2 (IN = PWM AND DIR)



SN7400: pin14=+5V, pin7=GND · all grounds common (Arduino + supply + TB6568KQ)

5. Physical connection list

From	To	Signal
Arduino D9 (PWM)	7400 pin 1 AND pin 9	PWM / speed
Arduino D12	7400 pin 2	DIR1
Arduino D13	7400 pin 10	DIR2
7400 pin 3	7400 pins 4 & 5 (jumper)	nA into inverter
7400 pin 8	7400 pins 12 & 13 (jumper)	nB into inverter
7400 pin 6	TB6568KQ pin 1 (IN1)	IN1
7400 pin 11	TB6568KQ pin 2 (IN2)	IN2
7400 pin 14 / pin 7	+5V / GND	Logic power
TB6568KQ pin 7 (VM)	+12V supply	Motor power
TB6568KQ pin 4 (GND)	Common ground	Ground

TB6568KQ pins 3 & 5	Motor leads	Motor out
All grounds	One node	Arduino+supply+driver+7400

6. Driving it from Arduino / Simulink

Action	D12	D13	D9 (PWM)
Forward	1	0	0-255 duty = speed
Reverse	0	1	0-255 duty = speed
Brake	1	1	255 (full)
Stop / coast	0	0	-

Bench cautions

- Tie off any unused 7400 input (all four gates are used here, so nothing floats).
- Motor power leads (red/black, +-12V) are NOT the encoder wires (5V) - swapping destroys the encoder.
- No common ground = no logic reference; the driver inputs float and behave randomly.
- This scheme is drive/coast (fast decay). For drive/brake (slow decay), hold one input high and PWM the other into the H,H brake state - different gate arrangement.
- Heat tab = GND; if bolted to a metal heat sink, the sink sits at ground potential.

7. Alternative design — 3 NAND gates, 2 Arduino pins

Same driver, fewer parts: use only **PWM (D9)** and a single direction bit **DIR (D12)**. The off-time between PWM pulses becomes a short brake instead of a coast, which removes the independent stop/coast state but gives smoother low-speed and position holding.

Logic (3 gates)

IN2 = NAND(PWM, DIR)

DIRb = NAND(DIR, DIR) # inverter, inputs tied

IN1 = NAND(PWM, DIRb)

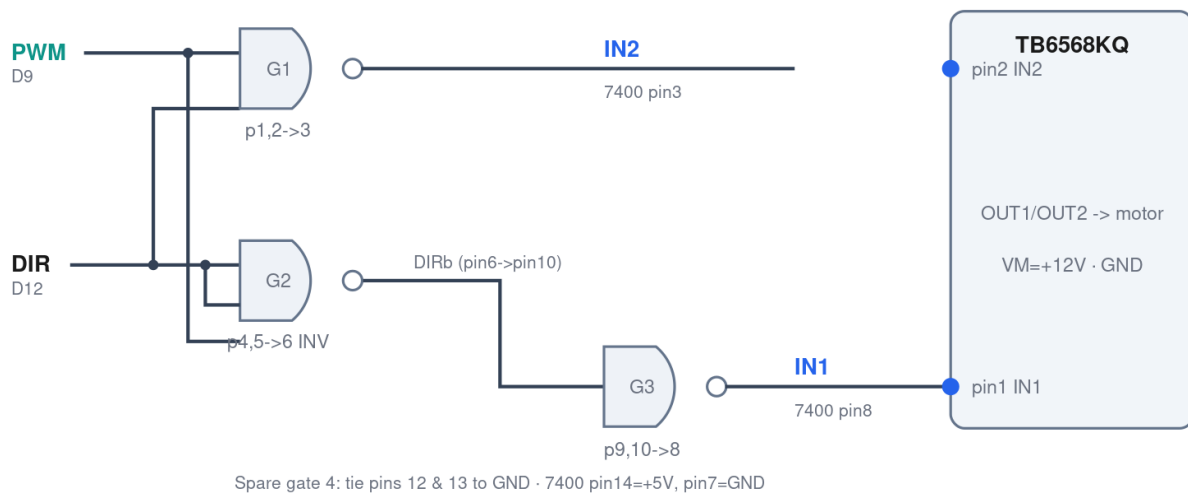
Behavior (verified against datasheet modes)

PWM (D9)	DIR (D12)	IN1	IN2	Mode
1	1	H	L	Drive CW
1	0	L	H	Drive CCW
0	x	H	H	Short brake

Duty = speed, DIR = direction. 0% duty holds the motor braked (stopped); 100% = full speed.

Gate wiring (one SN7400, 3 gates + 1 spare)

3 NAND gates · 2 pins -> IN1 / IN2



Connection list

From	To	Signal
Arduino D9 (PWM)	7400 pin 1 AND pin 9	PWM / speed
Arduino D12 (DIR)	7400 pins 2, 4 & 5	Direction bit
7400 pin 6 (DIRb)	7400 pin 10	inverted DIR
7400 pin 3	TB6568KQ pin 2 (IN2)	IN2
7400 pin 8	TB6568KQ pin 1 (IN1)	IN1
7400 pins 12 & 13	GND	spare gate 4 tie-off

7400 pin 14 / pin 7	+5V / GND	logic power
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8. Comparison — 3-gate vs 4-gate design

Aspect	4-gate design (Sec. 4)	3-gate design (Sec. 7)
NAND gates used	4 of 4 (full chip)	3 of 4 (1 spare)
Arduino pins	3 (PWM + DIR1 + DIR2)	2 (PWM + DIR)
Logic	IN1=PWM.DIR1, IN2=PWM.DIR2	IN2=NAND(P,D), IN1=NAND(P,D')
Decay / off-state	Drive / coast (fast decay)	Drive / brake (slow decay)
Duty = 0%	Motor coasts (freewheels)	Motor braked / held
Independent stop/coast	Yes (DIR1=DIR2=0)	No (off-time is brake)
Independent brake cmd	Yes (DIR1=DIR2=1)	Only as PWM off-time
Low-speed smoothness	Good	Better (active brake)
Position holding	Weaker (coasts)	Stronger (brakes)
Spare pins / gates	None	1 pin + 1 gate free
Best for	Full 4-mode control, coasting	Compact PID speed/position

Bottom line

Pick the **4-gate** version when you want the driver's full four-mode command set and a genuine coast state, and you can spare three Arduino pins. Pick the **3-gate** version when pins or gates are tight, or when you want tighter low-speed control and position holding for the PID labs — the active braking during PWM off-time is an advantage there. Both drive the TB6568KQ identically at the power stage; only the input logic and off-state behavior differ.